Model Performance Review

1 Coverage

- **Model outputs reviewed:** `OLS`, `OLS3`, `ENET`, `PLS`, `PCR`.
- **Prediction horizon reviewed:** 180 monthly out-of-sample periods from **2010-08-31** to **2025-07-31**.
- **Full-sample prediction rows per model:** 34,515.
- **Common artifact families used:** `r2`, `portfolio`, `benchmark`, `predictions`, `windows`, `importance`, `complexity`.

2 Key Takeaways

- **Best predictor on the headline squared-error metric:** `ENET`.
- **Best investable long-short model:** `ENET`.
- **Best long-only model:** `OLS`.
- **Most stable model across rolling windows:** `ENET`.
- **Weakest completed model in this batch:** `OLS3`.

2.1 My Takes

1. `ENET` is the strongest completed model overall. It has the best full-sample out-of-sample R^2, the strongest long-short net Sharpe after 30 bps cost, the highest benchmark information ratio, and the best cumulative long-short path.
2. `OLS` is not the best predictor on R^2, but it remains a strong long-only model. Its long-only net Sharpe is the highest in this completed batch even though its R^2 is negative. That means portfolio payoff and point-forecast fit are not the same thing in these runs.
3. `PLS` and `PCR` are credible middle-ground models. They do not beat `ENET`, but both deliver usable long-short Sharpe above 1 and materially better results than `OLS3`.
4. `OLS3` is materially weaker than the richer linear alternatives. In this dataset, the three-factor restricted specification is not capturing enough cross-sectional variation to compete.
5. The model ranking changes by objective. If the goal is **prediction fit**, `ENET` is best. If the goal is **long-only portfolio quality**, `OLS` is strongest among completed runs. If the goal is **best overall tradeable signal**, `ENET` is the best current choice.

3 Quick Comparison Table

Model	R2 OOS Full	R2 OOS Large	R2 OOS Small	Mean Rank IC	Long-only Sharpe	Long-short Sharpe	Benchmark IR
ENET	0.0044	0.0039	-0.0003	0.0785	0.9705	1.6960	0.2467
OLS	-0.0143	-0.0588	-0.0326	0.1044	1.2651	1.3013	0.0341
PCR	0.0025	0.0004	-0.0014	0.0545	0.8250	1.1464	0.0976
PLS	0.0008	-0.0019	-0.0034	0.0724	0.8423	1.1394	0.1408
OLS3	-0.0113	-0.0116	-0.0124	0.0306	0.3251	-0.0542	-0.2443

Note: *Although short selling is prohibited in Vietnam, this report still includes long-short results for cross-model comparison.*

Quick Notations and Formulas

Information coefficients (IC)

$$IC_t = \text{corr}(\hat{y}_t, y_t)$$

where, for each date t, $\hat{y}_t$ is the model score (or predicted return) across stocks and y_t is realized return across the same stocks. Then:

$$\text{Mean IC} = \frac{1}{T}\sum_{t=1}^{T} IC_t$$

Information Ratio (IR)

$$AR_t = R_t^{strat} - R_t^{bench}, \quad IR = \frac{\overline{AR}}{\sigma(AR)}$$

In this report, R_t^{bench} is the VN30 index (30 largest market-cap names at each test period, rebalanced every six months, equally weighted).

4 Prediction Quality

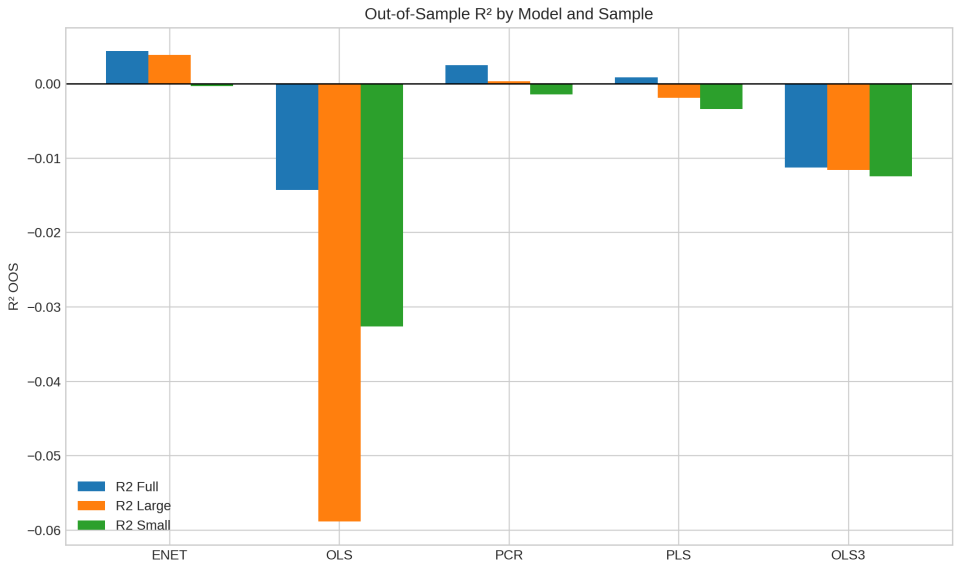

Figure 1: Out-of-sample R^2 by model

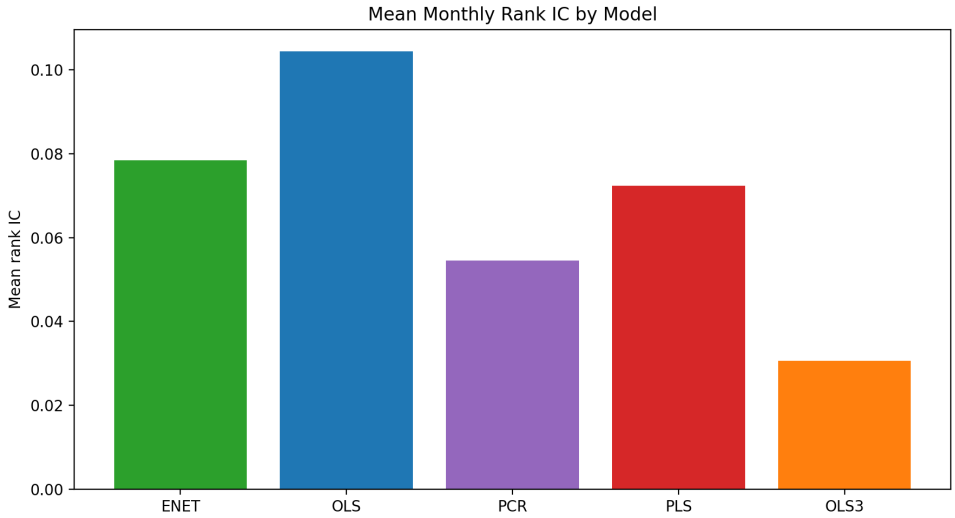

Figure 2: Mean rank IC by model

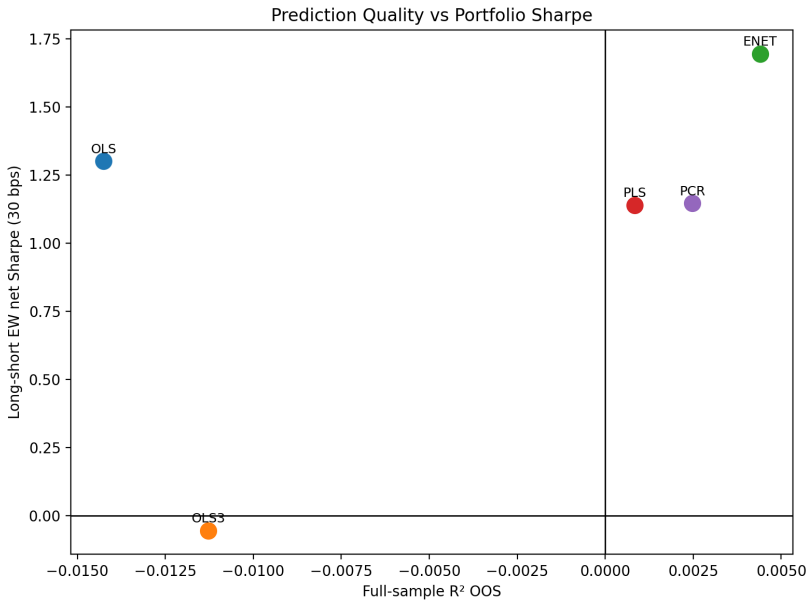

Figure 3: R^2 versus long-short Sharpe

4.1 Prediction Metrics Table

Model	R2 OOS Full	Pearson(y,yhat)	Spearman(y,yhat)	Mean Rank IC	Share Positive IC	Sign Accuracy
ENET	0.0044	0.0585	0.0535	0.0785	71.7%	51.5%
OLS	-0.0143	0.0629	0.0850	0.1044	76.7%	53.6%
PCR	0.0025	0.0529	0.0515	0.0545	68.9%	51.1%
PLS	0.0008	0.0551	0.0549	0.0724	70.0%	51.4%
OLS3	-0.0113	0.0182	0.0350	0.0306	56.7%	52.7%

Interpretation

- `ENET` is the only completed model with clearly positive R^2 in both full and large-firm samples.
- `PCR` is also positive on full and large samples, but its R^2 is weaker than `ENET`.
- `PLS` is barely positive on full sample and negative on large/small subsamples.
- `OLS` has the highest mean rank IC and highest sign accuracy, but still negative R^2.
- `OLS3` is weak on every predictive dimension reviewed here.

These results so far suggest

- For highest out-of-sample R^2, the answer is `ENET`.
- For monthly ranking ability, `OLS` looks better than its R^2 suggests.
- The safest interpretation is that `ENET` is the best balanced predictor, while `OLS` is noisier but sometimes economically stronger as a ranking signal.

5 Portfolio Performance

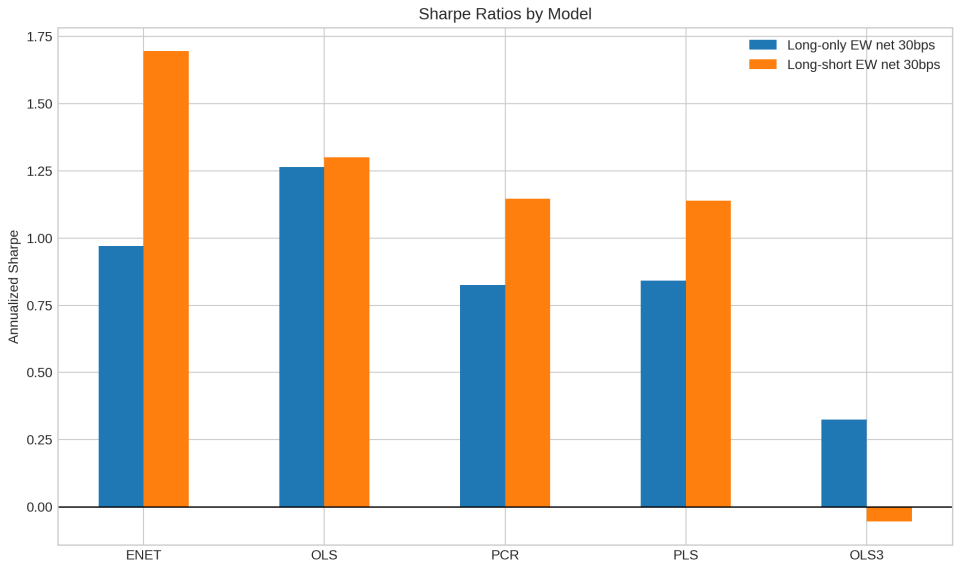

Figure 4: Sharpe by model

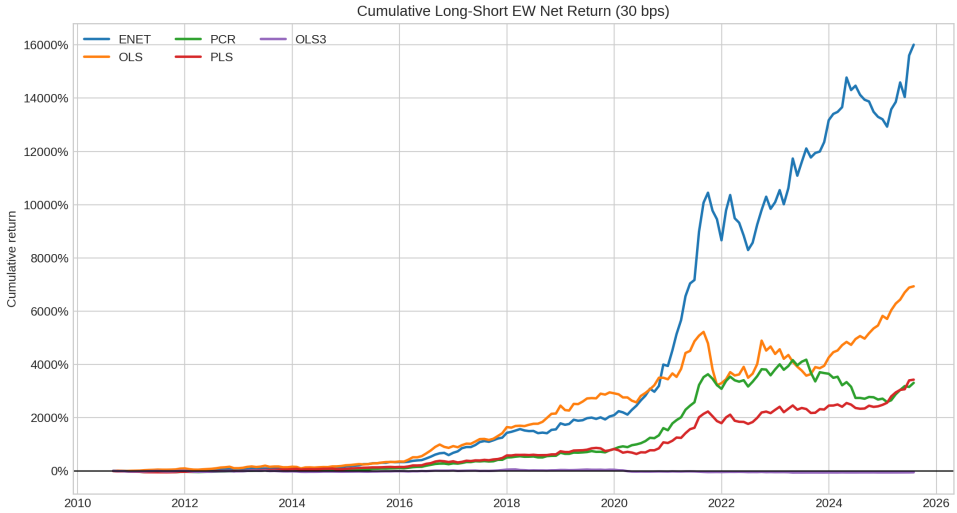

Figure 5: Cumulative long-short returns (net 30 bps)

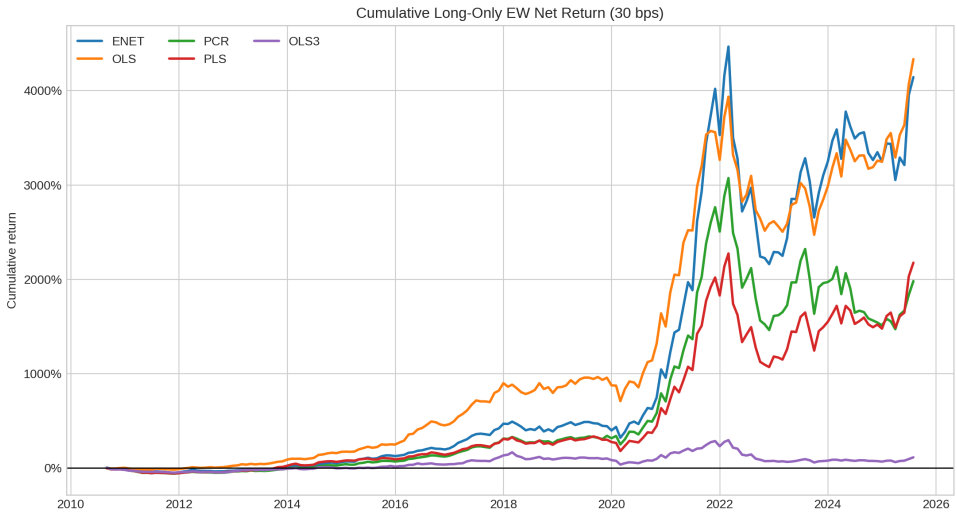

Figure 6: Cumulative long-only returns (net 30 bps)

5.1 Net 30 bps Portfolio Summary

Model	Long Ann Ret	Long Sharpe	Long Cum Ret (x)	Long Max DD	LS Ann Ret	LS Sharpe	LS Cum Ret (x)	LS Max DD	LS Mean Turnover
ENET	28.4%	0.970	41.434	-51.0%	40.3%	1.696	160.073	-26.2%	1.330
OLS	28.8%	1.265	43.343	-36.3%	32.8%	1.301	69.351	-37.6%	1.639
PCR	22.4%	0.825	19.805	-63.1%	26.5%	1.146	33.100	-50.1%	1.219
PLS	23.2%	0.842	21.752	-57.5%	26.8%	1.139	34.297	-48.1%	1.322
OLS3	5.1%	0.325	1.114	-60.2%	-4.5%	-0.054	-0.498	-74.8%	1.055

Interpretation

- `ENET` is decisively best on long-short portfolios: highest annualized return, highest Sharpe, and strongest cumulative return after 30 bps cost.
- `OLS` is the best long-only model in this completed set by net Sharpe and annualized return.
- `PLS` and `PCR` remain investable (long-short Sharpe above 1.1 after cost), but neither matches `ENET`.
- `OLS3` is not competitive; long-short is negative after cost and drawdown is the worst in the completed set.

6 Benchmark Relative Results

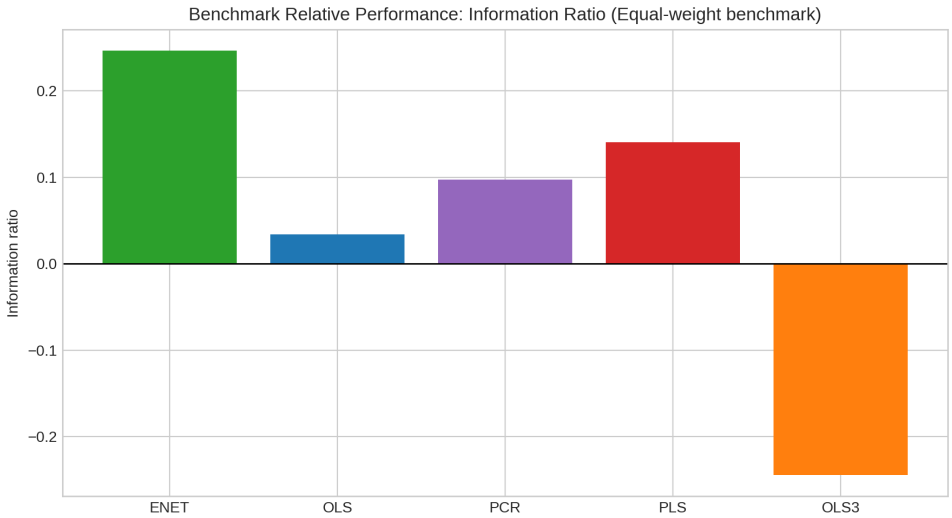

Figure 7: Benchmark information ratio (equal-weight)

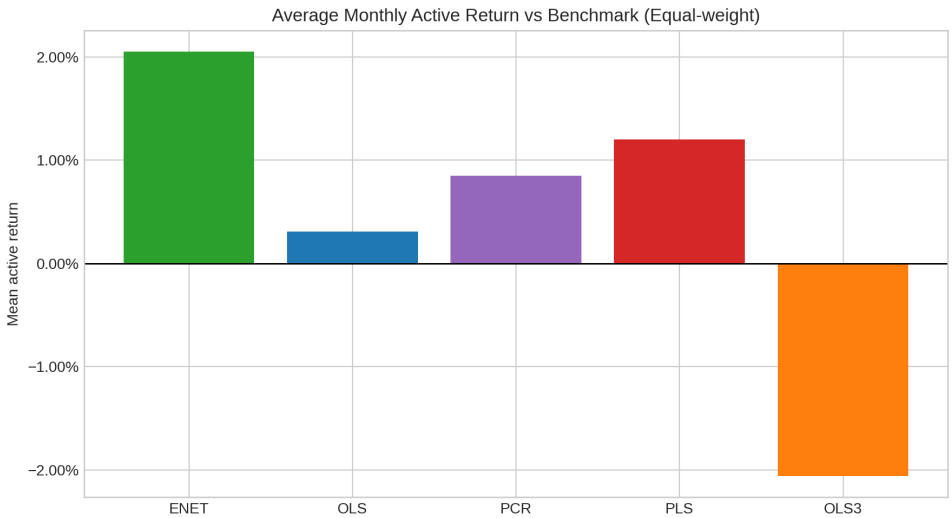

Figure 8: Benchmark active return (equal-weight)

Model	IR vs EW Benchmark	Mean Active Return	Corr vs Benchmark	IR vs VW Benchmark
ENET	0.247	2.05%	0.182	0.329
OLS	0.034	0.31%	-0.043	0.023
PCR	0.098	0.85%	0.090	0.107
PLS	0.141	1.20%	0.098	0.156
OLS3	-0.244	-2.06%	0.074	-0.223

Interpretation

- `ENET` is the strongest benchmark-relative strategy by a clear margin.
- `PLS` and `PCR` are second tier but still positive.
- `OLS` is only slightly positive against the equal-weight benchmark.
- `OLS3` is negative against both equal-weight and value-weight benchmark comparisons.

7 Rolling Window Stability

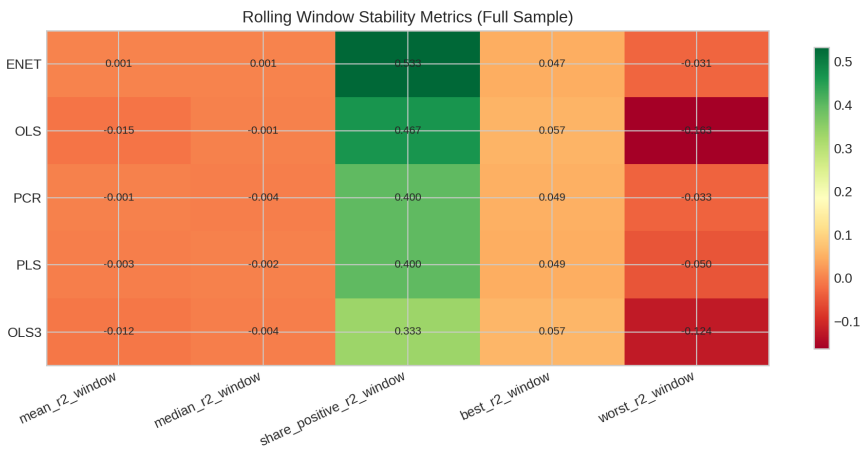

Figure 9: Rolling-window stability heatmap

Model	Mean Window R2	Median Window R2	Share Positive Windows	Best Window R2	Worst Window R2
OLS	-0.0149	-0.0011	46.7%	0.0565	-0.1631
OLS3	-0.0115	-0.0038	33.3%	0.0574	-0.1242
ENET	0.0013	0.0009	53.3%	0.0467	-0.0307
PLS	-0.0030	-0.0021	40.0%	0.0486	-0.0499
PCR	-0.0007	-0.0043	40.0%	0.0492	-0.0326

Interpretation

- `ENET` is the only completed model with positive mean window R^2 and more than half of windows positive.
- `OLS` has the best single window but much worse downside window risk than `ENET`.
- `PCR` and `PLS` are middling and slightly negative on average window R^2, but less erratic than `OLS`.
- `OLS3` is consistently weak.

8 Turnover Cost Sensitivity

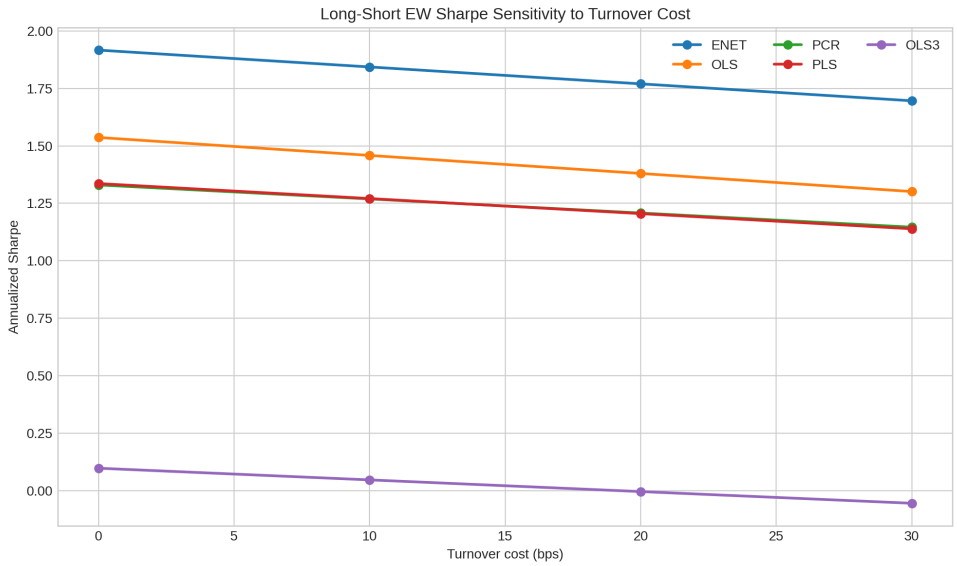

Figure 10: Cost sensitivity of long-short Sharpe

Model	LS Sharpe 0bps	LS Sharpe 10bps	LS Sharpe 20bps	LS Sharpe 30bps	Long Sharpe 30bps
OLS	1.536	1.458	1.380	1.301	1.265
OLS3	0.098	0.047	-0.003	-0.054	0.325
ENET	1.916	1.843	1.770	1.696	0.970
PLS	1.336	1.271	1.205	1.139	0.842
PCR	1.330	1.269	1.208	1.146	0.825

Interpretation

- `ENET` remains best on long-short Sharpe even after 30 bps turnover cost.
- `OLS` starts with strong gross results but degrades faster than `ENET` once costs are applied.
- `PLS` and `PCR` are moderately resilient to cost, but still clearly below `ENET`.
- `OLS3` is effectively non-viable once cost is introduced.

9 Feature Importance and Model Shape

Figure 11: Feature-importance heatmap

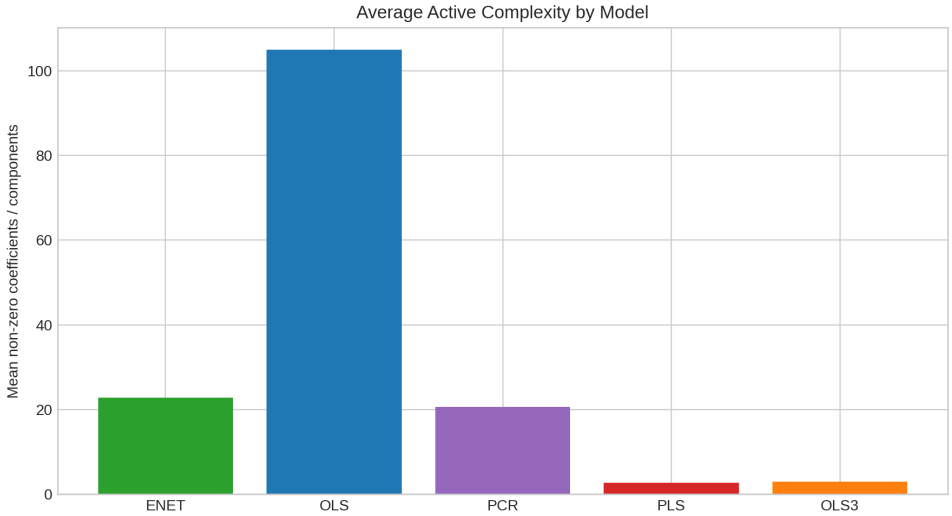

Figure 12: Complexity by model family

9.1 Top Features by Model

Model	Top 1	Top 2	Top 3	Top 4	Top 5
ENET	mom1m (0.789)	Shares_Out (0.566)	cfp (0.039)	ROE_Reported (0.000)	EBITDA (0.000)
OLS	FCF (0.120)	US_Bond_10Y (0.079)	std_turn (0.077)	mom6m (0.065)	Hong_Kong_Index (0.061)
PCR	Oper_CF (0.075)	agr (0.075)	dollar_vol (0.074)	Cur_Assets (0.074)	Inventory_BS (0.074)
PLS	P/B (0.709)	Price (0.107)	Shares_Out (0.061)	FCF (0.038)	std_turn (0.035)
OLS3	me (1.215)	be_me (-0.004)	ret_12_1 (-0.212)		

9.2 Complexity Summary

Model	Mean Validation Score	Mean Active Complexity	Min Active Complexity	Max Active Complexity
OLS	0.298	105.000	105.000	105.000
OLS3	0.141	3.000	3.000	3.000
PLS	0.141	2.667	1.000	6.000
ENET	0.141	22.867	5.000	54.000
PCR	0.140	20.667	1.000	49.000

Extra Notes

How Feature Importance Was Calculated

First, compute a baseline fit score on `base` with the same `R2OOS` function used elsewhere:

$$R^2_{\text{base}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i y_i^2}$$

Then, for each feature f, set that feature to 0.0 in `train`, `validation`, and `base`, refit, predict, and recompute:

$$\Delta R^2_f = R^2_{\text{base}} - R^2_{f=0}$$

Finally, normalize into variable-importance shares:

$$\text{var_imp}_f = \frac{\Delta R^2_f}{\sum_j \Delta R^2_j}$$

How Model Complexity Was Calculated

For each rolling window, the model returns:

- `best_params`: hyperparameters chosen on the validation set,
- `best_score`: validation objective value of the selected specification,
- `complexity`: a compact model-shape summary.

Complexity by model family

- **OLS / OLS3**: implemented with `HuberRegressor`. `best_score` is validation RMSE and complexity uses `n_nonzero_coef`. For `OLS3`, this remains 3 because features are fixed to `[me, be_me, ret_12_1]`.

- **ENET**: `best_score` is validation RMSE; complexity includes `best_alpha`, `best_l1_ratio`, and `n_nonzero_coef`.
- **PLS**: `best_score` is validation RMSE; complexity uses `n_components`.
- **PCR**: `best_score` is validation RMSE; complexity uses `n_components`.
- **GBRT**: `best_score` is validation Huber loss; complexity records max depth, min samples split, min samples leaf, and learning rate.
- **RF**: `best_score` is validation MSE; complexity records max depth and max features.